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Expert Network Flexible Foam Q1 2024

Summary of Global Purchasing Meeting with Huntsman, Dow and Wanhua

Autoneum. Mastering sound and heat.

Non-Disclosure Statement

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Please note that the majority of the following information has been shared under NDA with us.

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Agenda

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- Global Purchasing Meeting with Dow, Huntsman and Wanhua
  - Non-Disclosure statement
  - Autoneum participants at global purchasing meeting
  - Sustainability Initiatives at Huntsman
  - Other topics with Huntsman
  - Sustainability Initiatives at Dow
  - Other topics with Dow
  - Sustainability Initiatives at Wanhua
  - Other topics with Wanhua
  - Status of new system introduction of Wanhua in China

Autoneum participants at global purchasing meeting

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Daniela Riera

Matthew Miller

John Zhu

Hermann de Ciutiis

Jean-Yves Aubert

Sonsoles de Olano

Ciro Gaudino

Maurizio Mantovani

James Taylor

Philippe Godano

Christopher Noakes

Luca Marotta

Wolfgang Tenbrock

John Trumble

Erich Jeckle

Carbon Footprint Reduction - Huntsman

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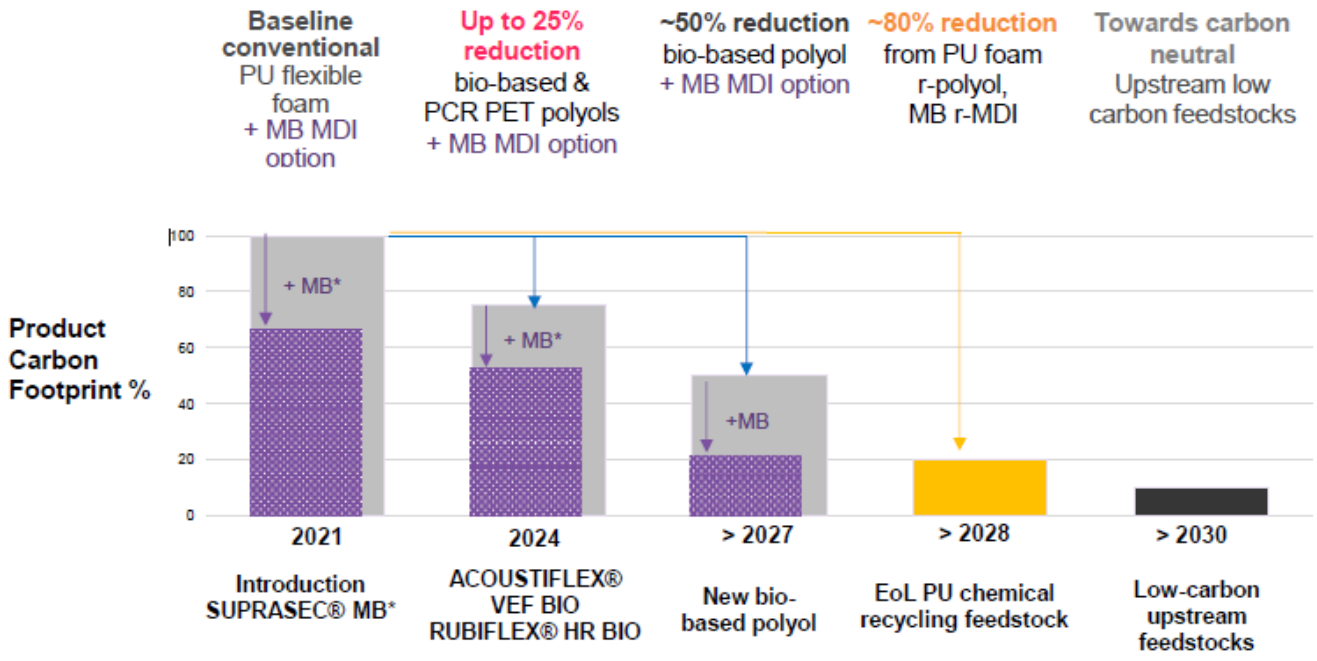
Our Roadmap Toward Net Zero – Cradle to Gate

Our ambition for carbon footprint reduction – from raw materials to customer delivery



Legend of terms
EoL – end-of-life
GWP – global-warming potential emission
LCA – life-cycle analysis
MB – mass-balance
PCR – post-consumer recycled
PIR – post-industrial recycled (scrap)
r- – indicates recycled

Functional Unit 1 kg of foam
Value at 100% as reference, representing in %s the:
GWP_{100} including biogenic CO_2 (kg CO_2 eq/kg)

* additional 33% attributed reduction using mass-balanced bio-circular benzene
** PU EoL waste input based on cut-off rule

Comment:

- CFR of biocircular benzene not necessarily recognized +1500 EUR/ton
- PCR PET trials ongoing in R&T, performance insufficient so far
- bio-based polyols bear the risk of a monopolistic dependence
- EoL supply chain, separation and process to be developed

Carbon Footprint Reduction – Huntsman EoL foam

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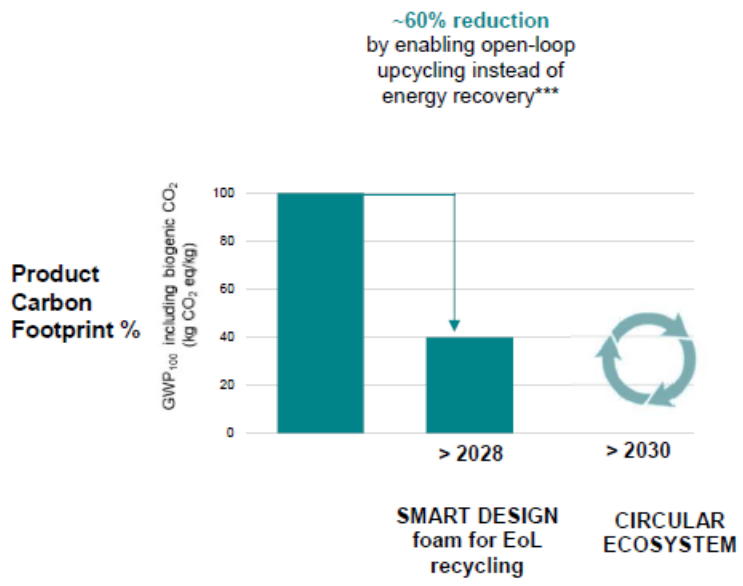
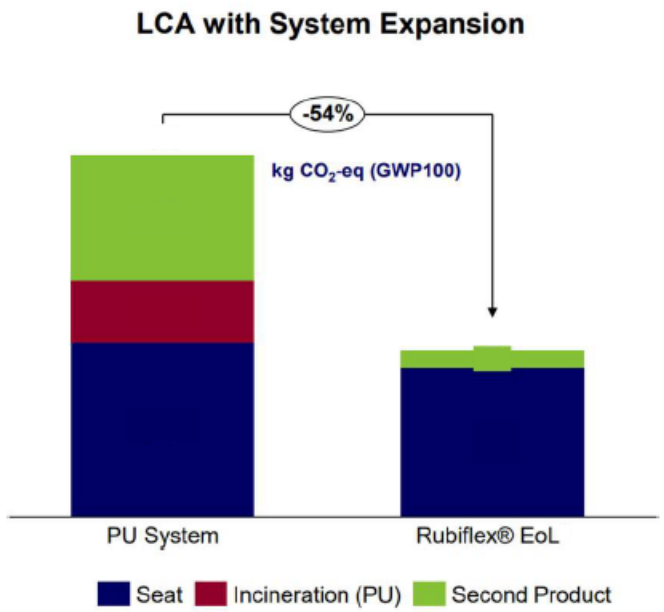
Our Roadmap Toward Net Zero – Cradle to Grave

Our ambition for carbon footprint reduction – material full life into second life





Functional unit: 1 seating foam, including EoL scenario

***(LCA study based on system expansion)

Comment:

- need to understand the acceptance of system expansion by our customers
- need to determine how to share the advantage between the first product cycles

Sustainability Initiatives – Huntsman

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Sustainability Initiatives

Leading circularity in polyurethane foams & composites

100% RECYCLABLE FOAM & MONO-MATERIAL DESIGNS

- Design mono-material heavy-layer and foam
- Molded with existing process or slabstock for thermoforming
- 100% recycled back into automotive parts
- **Discuss:** designing multi-layer, mono-material acoustic solution

CIRCULAR MATERIALS – UP TO 25% CO_2 FOOTPRINT REDUCTION*

- Bio-renewable & post-consumer recycled PET polyol
- Ready to trial for acoustic foams and composites
- **Discuss:** global demand to drive scale-up of production

ADVANCED CHEMICAL RECYCLING

- Process yields recycled polyol and MDI for higher value
- In lab - ~60% recycled in new foams
- Challenge: source of EOL foam, economics of dismantling
- **Discuss:** participation in pilot and scrap foam sourcing

BIO-CIRCULAR MDI

- Rotterdam manufacturing ISCC+ certified
- Enables up to 30% CFR* in foam
- Does not require new formulation
- **Discuss:** certification justification for APAC & NA

*Source: Preliminary LCA results - cradle-to-Huntsman gate. Component for manufacturing of polyurethane systems. LCA method GWP $_{100}$ including biogenic CO_2 CML2016 Software: GaBi 2021-2

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separate, sort, shred, supply, compress to new product

tear strength is low, color not constant, trials ongoing

expensive solutions

Advanced Chemical Recycling

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Advanced Chemical Recycling

Promising chemistry - cost-effectiveness a hurdle

- Split-phase hydroglycolysis in lab **delivers recycled polyol and MDI**
- Hurdle – recycling costs of dismantling/separation of foam and infrastructure
- **Pilot with right partner could validate and launch initial program from scrap foam**
- Need EOL foam to eventually scale up due to limited scrap and European EOL vehicle proposal requirement for post-consumer content

Polyols Physical Properties Overview Virgin vs Recycled			
Property	Measure	Virgin	Recycled
Density	kg/m^3	64.1	66.1
CLD 40%	kPa	9.3	9.0
Hysteresis 70%	%	26.0	23.7
Tear strength (mean)	N/cm	2.65	2.29

Typical properties can vary depending on local circumstances and application. These properties are not part of the specifications of the polyols.

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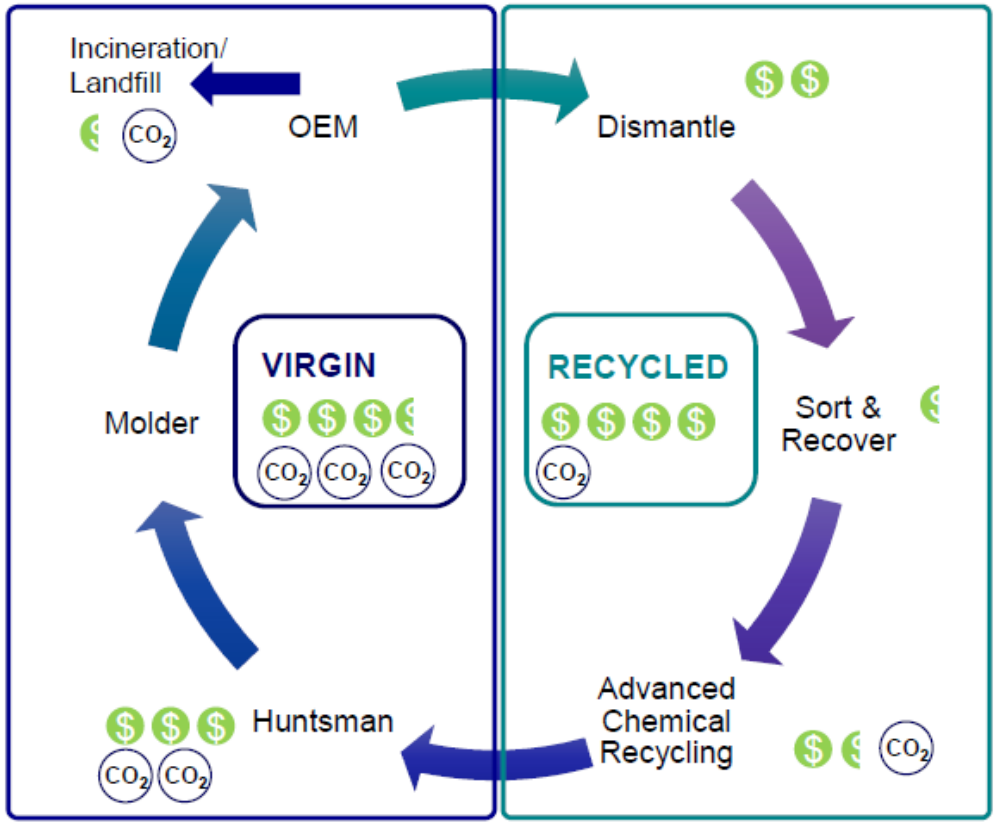

Comment:

- other companies use the term “advanced chemical recycling” as well but might use it for a different process with potentially higher CO_2 load – it depends how far down you split the initial molecule. Huntsman separates the new liquid in 2 phases and will use one phase as such.

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Other topics with Huntsman

High Temperature (180°C, 7 days) stable foam for engine bay

Low modulus foam benefits:

- High heat resistance (7 days at 180°C)
- Non burning
- Appealing visual aspect
- Bio content
- UL94 V0
- Standard processing applicable

ACOUSTIFLEX® HT
Engine cover and beauty cover solution – ready to trial

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Comment:

- Tested in Ellzee by Erich Jeckle:
  Foam failed at 180°C for 30 minutes
- Works at 170°C for 30 minutes

- Requirement : 180°C for 5 h

Heated floormats in combination with Huntsman's Elastomers portfolio

- fully recyclable
- TPU
- Follow up is planned

Theta-Cell S12 introduction everywhere

- Successfully switched plant Chocen including moving to new catalyst
- What is the status for China / Oregon?

Global Purchasing Meeting

Other topics with Huntsman

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VE Foam systems next step

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Global Formulations: ACOUSTIFLEX® VEF

Assured supply, standardization possible

CURRENT SUPPLY PER SITE

SITE	TE	LE	ULE
Shenyang Texi CN	Running		Ready
Bloomsburg USA		Running	Ready
Aiken USA		Running	Ready
San Louis Potosi MX		Running	Ready
Bor CZ		Running	Ready
Gundernhausen, DE		Running	Ready
Elzee, DE	Running		Ready

Benefits of moving to Ultra Low Emission:

- Increase competitiveness
- Potential removal of processing steps
- Increase in processing flexibility

TE – Typical emissions, LE – Low emissions, ULE, Ultra low emissions

Discuss: We can standardize and it is available, what is our next step?

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Global Formulations: ACOUSTIFLEX® HR

Assured supply, standardization possible

CURRENT SUPPLY PER SITE

SITE	TE	LE	ULE
Genk BE	Running		Ready
Bloomsburg USA		Running	Ready
Halesowen UK		Running	Ready
Stoke on Trent UK		Running	Ready
Katowice PL		Running	Ready
Komarom, HU		Running	Ready

Benefits of moving to Ultra Low Emission:

- Increase competitiveness
- Potential removal of processing steps
- Increase in processing flexibility

TE – Typical emissions, LE – Low emissions, ULE, Ultra low emissions

Discuss: We can standardize and it is available, what is our next step?

Comment:

- need to do our homework:
- What is the real emission status of the different foam systems?
- What is required where for future RFQ?
- How do we tackle it considering self-formulation activities?

Sustainability initiatives – Dow

Reduce carbon emission at e.g. Terneuzen (NL) site (polyol production)

- Hydrogen plant, CO_2 capture and storage, e-cracking using renewable electricity
- 40% reduction by 2030 -> 25% reduction of CO_2 footprint on polyol
- Reduction is not evenly distributed to all products, but you need to pay for it according to the attributed reduction level (500 EUR/ton)
- Products marketed as «DEC» for decarbonization

Advanced chemical recycling (existing and available, all European plants certified)

- Recycled and biobased feedstock to replace virgin oil
- Attributed to single products via mass-balance, you have to pay according to the attributed recycling content
- Circularity levels of 60% for iso and 75% for polyol is achievable, marketed as “CIR” with a 500 EUR/ton surcharge

Renuva polyol (existing and available, collection agreements in various countries)

- Polyol from mattress recycling, up to 25% in slabstock polyol -> 6% EoL recycling content in new product

Other PU Foam topics with Dow

VE Foam (ready to trial in Mex, but price issue)

- System SPECFLEX 1064 Polyol / SPECFLEX NE 562 Isocyanate is commercial
- Applied density 60 kg/m3 (down to 10mm in flat tool without defects)
- Demold time 60-75 s on carpets
- Passing emission & odor requirements of BMW and Mercedes in combination with RA KVS 5211 and ACMOS 37-9529

Theta-Cell

- Sustainable (Renuva) polyol can be used
- Replacing catalyst in Theta-Cell for legislation reason
- Trying a new FR in Theta-Cell which coud lead to lower usage of graphite

Battery

Differnet options for battery, want to understand the application and ist value before proposing a system and giving a price

Propose Voratron EP500 for battery filler foam

PUSH

- Propose SPECFLEX ACTIVE to achieve faster demolding time. Compared to VORACTIVE it provides gel activitiy in addition to blow

Sustainability initiatives – Wanhua

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Bio based polyether (based on Castor oil)

- Offer 3 different grades of biobased-polyol with 37%, 51% and 91% biobased carbon
- Example shown with 15% bio-based raw material content for acoustic foam
- CO_2 emission reduction 30-50%, at least 40% higher price

Comparison of Physical properties

Items	Petro-based foam	Bio-based foam
Bio-based foam content, %	0	15
Density, kg/m³	60	60
Tensile strength, kPa	234	155
Elongation at break, %	95.5	96
Tear strength, N/cm	2.31	2.2
CLD40%, kPa	6.49	5.63

Comparison of emission

products	PA	ACE	Acrolein	BEN	MB	EB	SM	DMB	TVOC
Petro-based	33	60	10	ND	ND	ND	ND	ND	477
Bio-based	35	70	18	ND	10	ND	ND	ND	369

Test method: 10L bag

5

Decarbonization

- sea-based and land-based solar PV, wind power, nuclear energy
- Target to achieve > 50% proportion of green electricity -> annual reduction of 37M tons CO2

Reuse Heat waste

- Use of excess process heat for civil heating -> annual reduction of 4.9M tons CO2

Develop and improve CCUS (carbon capture utilization and storage) platform

Carbon Goals Wanhua

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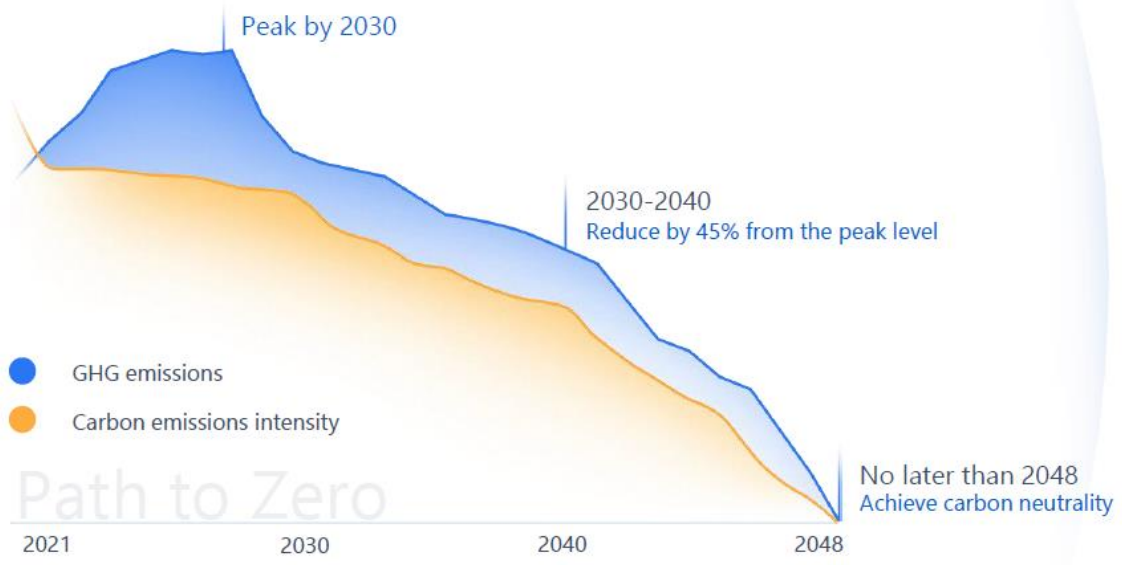
Dual Carbon Goals

WANHUA

Wanhua Chemical is committed to achieving peak carbon no later than 2030 and carbon neutrality by 2048.



*Scope 1+2

Key emission reduction measures

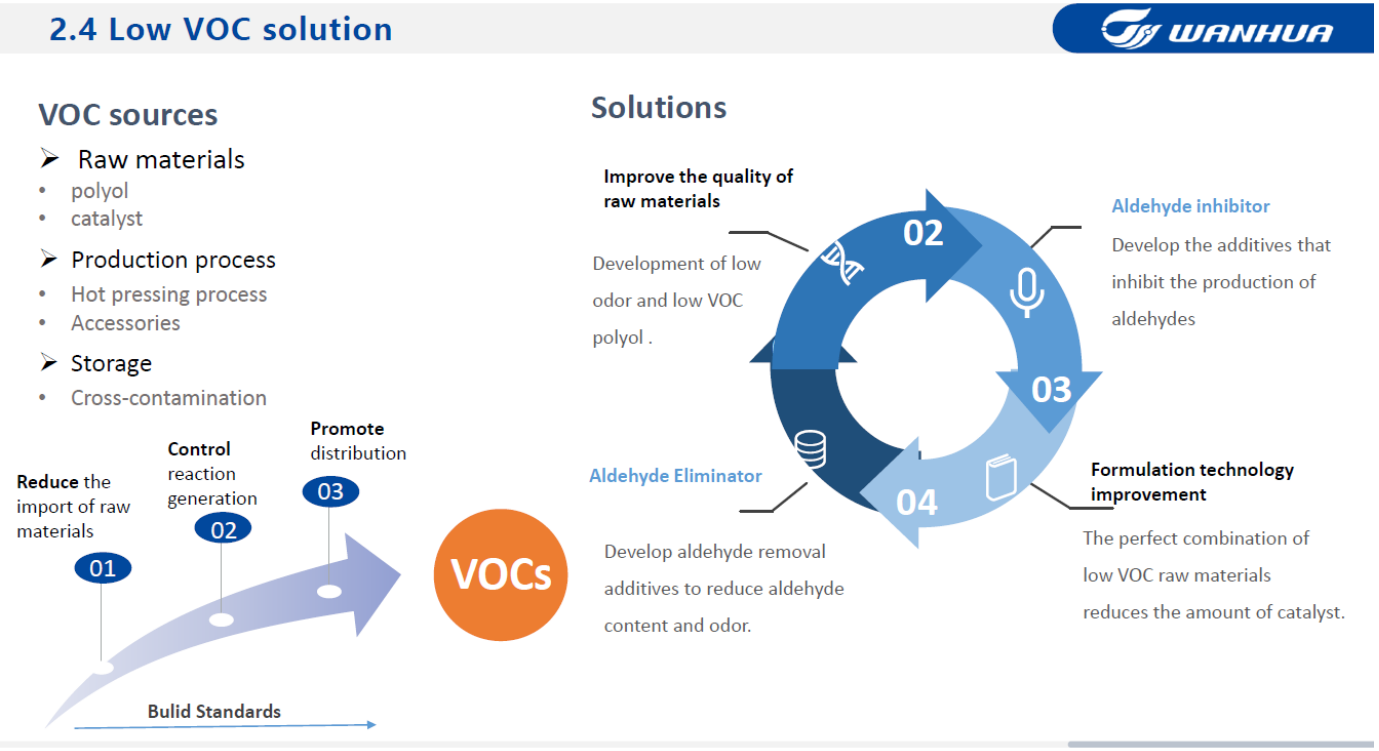
Global supply situation – Wanhua

- 5 production sites in China
- 1 MDI and TDI production site in Europe
- EU system house in 2025, 10kt flexible foam, 50kt rigid foam
- Mexico system house in 2027 (in Monterey), 10kt MDI variants, 10kt form flex polyol, 50kt rigid foam

Other topics with Wanhua

Low VOC solution for foam

- Claim to be seen as the benchmark for aldehyde reduction in China



Other topics – Wanhua

Battery: several options to be discussed within other Autoneum teams

- Wanhua is in direct contact with battery manufacturers

Innovation

WANHUA

- **Wanhua invests lots of resources to research and develop innovative products . Innovation is the main driving force to Wanhua's development.**

WANHUA CATL 宁德时代

Cooperate with Catl to develop battery solutions

Upper cover of battery case

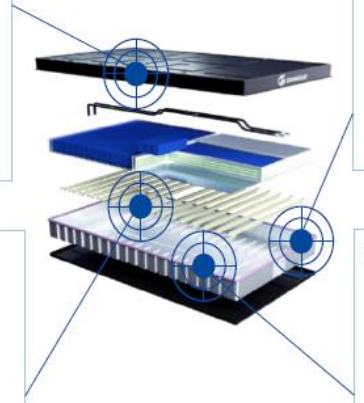
- **Market demand**: 10000t/a
- **Current progress**:

Mass production in Huangyang;Small batch in shansheng;

Passed the trial in Weishen and Feirongda.

- **Plan in 2023**:

Mass production in Feirongda,Guoen and Weisheng;



Bottom protective plate

- **Market demand**: 50000t/a
- **Current progress**: Passed UL94 (V0 grade) ;passed thermal runaway test Completed the package test of A sample
- **Plan in 2023**: **Optimize the production process**

Anti-hit bracket

- **Market demand**: 3000t/a（Kirin 3.0 battery pack）
- **Current progress**: Passed verification; mass production in huangyang , kalai , guoen ;
- **Plan in 2023**: Stablely applied in Kirin 3.0 battery.

Bumper block

- **Market demand**: 20000t/a
- **Current progress**: Fulfill CAE analysis , passed UL94 (V0 grade); Completed the process development and preparation of A sample
- **Plan in 2023**: Produce 10,000 battery packs, Installation verification.

Wanhua at Autoneum in China

WANHUA

The Supply Share of Wanhua in AUTONEUM plants in China

Shenyang Plant

- 01 Volvo K431 — Carpet Systems

- 02 Volvo L541 — Carpet Systems

Taicang Plant

- 01 Volvo XC40 — Carpet Systems and Inner Dashes

- 02 CJLR L551 — Inner Dashes

- 03 Honda — Inner Dashes

HONDA

Chongqing Plant

- 01 Volvo XC60 — Carpet Systems and Inner Dashes

Pinghu Plant

- 01 Polestar 2 — Carpet Systems and Inner Dashes

- 02 M 3 — Carpet Systems

Guangzhou Plant

- 01 Xiaopeng E38 — Carpet Systems

- 02 GAC A39 — Carpet Systems

Projects in Development

- 01 BMW G18/G28 — Wheelhouse liners
- 02 BMW — Light weight foam, Carpet Systems
- 03 VEF+ — Carpet Systems

AUTONEUM Shenyang Plant

- 04 Volvo 732K/Polestar 6 — Inner Dashes and Wheelhouse liners — Chongqing Plant
- 05 BBAC MMA 252 — Carpet Systems — Tianjin Plant

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